FUTURE OF WORK · GOOGLE AI BUILDER CUP

# ANVIL

## Governance for autonomous spending agents.

A human signs what an agent may spend. Anvil checks every action against it, blocks drift, and makes the agent fix itself.

THE PROBLEM

# Agents already spend money. Nothing checks they stayed inside what a human approved.

- Travel & expense agents book flights and hotels on their own today.
- Rules engines only see numbers: a price under the cap passes, even when the **cabin class**, **refundability** or **intent** is wrong.
- When something goes wrong, there is no tamper-evident record of who authorized what.

**Example: "within budget" but out of policy**

Agent books **business class** on a non-preferred airline, $590 against a $600 cap. Every budget rule passes. The manager never authorized it.

[ADD: one sourced statistic from the project brief on agentic spend / T&E leakage, with source]

WHAT ANVIL IS

# A signed mandate in, a verdict and an audit trail out.

**1 · Signed mandate**
Manager's limits as an AP2 Intent Mandate (HMAC-signed, verified before every action)

→

**2 · Agents over A2A**
Flight, hotel, transport agents discover suppliers via Agent Cards, JSON-RPC

→

**3 · Anvil auditor**
Gemini judges the action against the intent of the mandate; structured JSON

→

**4 · Decide**
**MATCH** allow + Cart Mandate
**REVIEW** one-tap re-consent
**DRIFT** block + explain

→

**5 · Provenance**
Hash-chained log with delegation chain, live on the dashboard

Runs on Cloud Run (orchestrator + supplier agents), Gemini on Vertex AI, Firestore, Firebase Hosting.

THE TWIST

# Anvil doesn't just say no. The agent corrects itself.

BLOCKED

**Emirates · business · $590**

"Flight class 'business' does not match mandated 'economy' class."

→

SELF-CORRECTED · MATCH

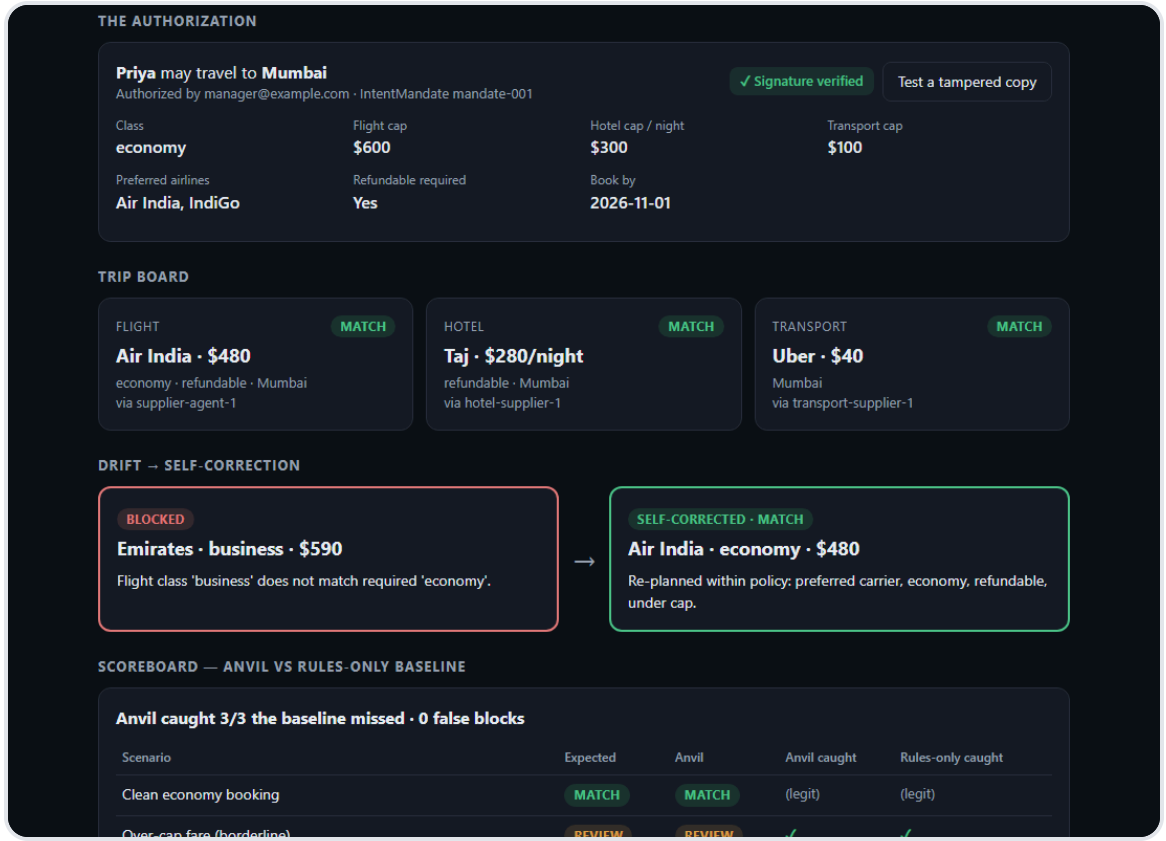
**Air India · economy · $480**

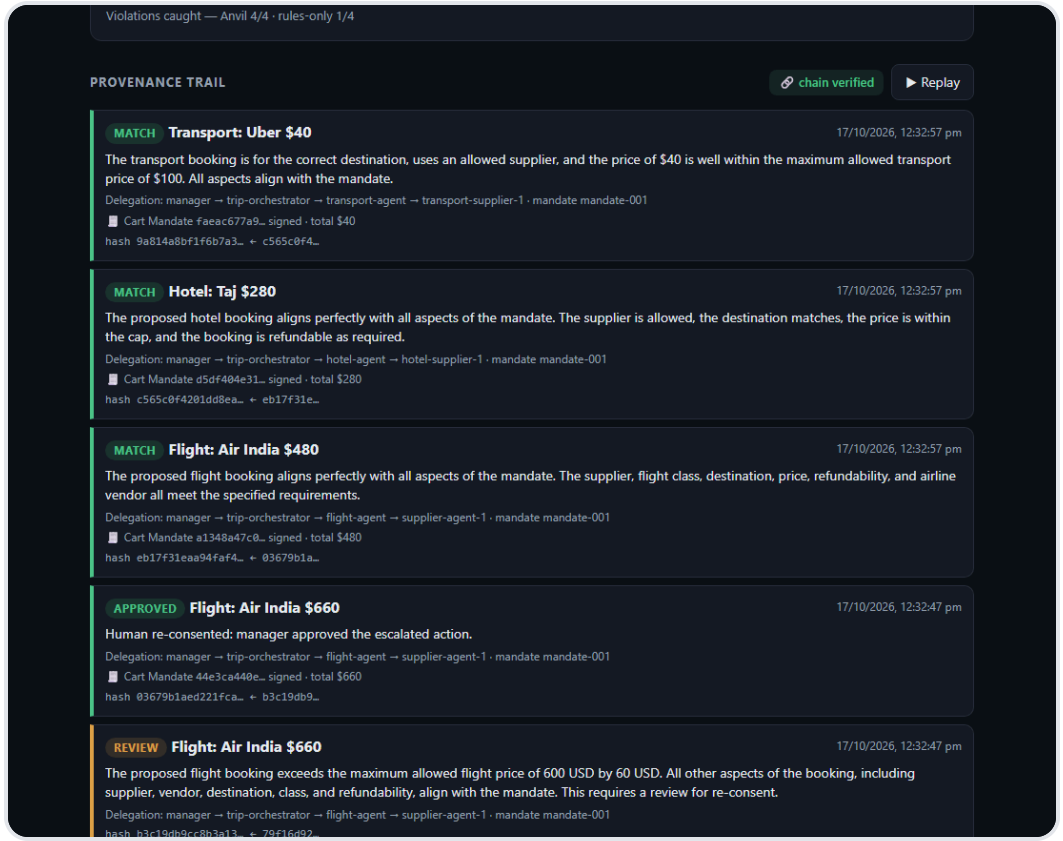
Re-planned within policy: preferred carrier, economy, refundable, under cap.

- Intent-drift reasoning: cabin class, refundability, and a booking **split into two charges to dodge an approval cap**.
- Borderline cases are REVIEW, not blocks: escalated for one-tap manager approval.

LIVE DEMO

# One mandate. Three agents. One trail.




Live on Google Cloud. Left: signed mandate, trip board, drift → self-correction. Right: hash-chained trail with delegation and Cart Mandates.

THE PROOF

# Measured, not asserted: same actions, two systems.

| SCENARIO | EXPECTED | ANVIL | RULES-ONLY CAUGHT |
|---|---|---|---|
| Clean economy booking | MATCH | MATCH | (legit) |
| Over-cap fare | REVIEW | REVIEW | ✓ |
| Non-refundable fare | REVIEW | REVIEW | ✗ |
| Business-class upgrade | DRIFT | DRIFT | ✗ |
| Split charge to dodge cap | DRIFT | DRIFT | ✗ |

**4/4**
violations caught by Anvil

**1/4**
caught by a rules-only baseline

**0**
false blocks on legitimate actions

Identical verdicts across repeated runs (temperature 0). Reproduce: `python -m anvil.eval` or `/api/eval`.

GOOGLE STACK

# Built end to end on Google's agent stack.

### Gemini · Vertex AI

The drift auditor: structured JSON verdicts at temperature 0, with retries and a cached-verdict fallback.

### A2A

Agent Cards + discovery; Message/Task over JSON-RPC between buyer and supplier agents.

### AP2

Signed Intent Mandate (human authorization) and Cart Mandate (per-purchase), verified before every action.

### Cloud Run

Two services: orchestrator + guard + API, and the A2A supplier agents.

### Firestore

Mandates and the hash-chained, tamper-evident provenance log.

### Firebase Hosting

Live dashboard with a real-time Firestore listener.

Hand-rolled, faithful to the A2A / AP2 core objects (not the official SDKs; HMAC signatures, not Verifiable Credentials).

HOW IT BROADENS

# Beachhead: travel & expense. Destination: the control plane for agent spending.

**Today · T&E**

Flights, hotels, ground transport governed by one signed mandate.

**Next · Cloud & SaaS**

Agents that provision infrastructure and buy seats, checked against the same kind of mandate.

**Then · Procurement**

Vendor selection and purchase orders, with delegation chains and approvals built in.

Same engine, same three verdicts, same tamper-evident trail. Only the mandate and the supplier agents change.

WHY NOW · IMPACT

# Agents are live today, and spend leaks without a guard.

[ADD]

Agentic travel/spend already in production (name the real products + source from the brief)

[ADD]

Size of T&E leakage / out-of-policy spend (figure + source from the brief)

[ADD]

Time saved by one-tap re-consent vs manual review (only if measured)

Anvil gives finance and compliance teams what they will need before they let agents spend: proof of authorization, a block on drift, and a trail an auditor can verify.

TEAM · ASK

# Let agents spend, with a human's limits intact.

**Team**

[ADD: team name, members and roles]

**Ask**

[ADD: what you want from the judges / next step]

Live: https://dazzling-lodge-511607-m0.web.app
Code: https://github.com/Adhi1402/anvil